

DBMS Viva + Practical Master Notes (SPPU SE 2024 Pattern)

1. What is DBMS?

DBMS (Database Management System) is software used to create, store, manage and retrieve data efficiently.

2. What is Database?

A database is an organized collection of related data stored electronically.

3. What is Table?

A table is a collection of rows and columns used to store data.

4. What is Tuple?

A tuple is a single row or record in a table.

5. What is Attribute?

An attribute is a column in a table.

6. What is SQL?

SQL is Structured Query Language used to manage databases.

7. What is DDL?

DDL defines database structure. Commands: CREATE, ALTER, DROP.

8. What is DML?

DML modifies data. Commands: INSERT, UPDATE, DELETE.

9. What is DCL?

DCL controls permissions. Commands: GRANT, REVOKE.

10. What is TCL?

TCL manages transactions. Commands: COMMIT, ROLLBACK.

11. CRUD Operations

Create, Read, Update, Delete.

12. Primary Key

Uniquely identifies each record in a table.

13. Foreign Key

Connects two tables using primary key reference.

14. Super Key

Uniquely identifies records.

15. Candidate Key

Minimal super key.

16. Alternate Key

Candidate key not selected as primary key.

17. What is JOIN?

Combines rows from multiple tables.

18. INNER JOIN

Returns matching rows from both tables.

19. LEFT JOIN

Returns all rows from left table and matching rows from right table.

20. RIGHT JOIN

Returns all rows from right table and matching rows from left table.

21. FULL JOIN

Returns all rows from both tables.

22. Normalization

Organizing data to reduce redundancy.

23. 1NF

All attributes contain atomic values.

24. 2NF

No partial dependency.

25. 3NF

No transitive dependency.

26. View

A virtual table created from SQL query.

27. Transaction

Sequence of operations treated as single unit.

28. ACID Properties

Atomicity, Consistency, Isolation, Durability.

29. Cursor

Processes rows one by one.

30. Types of Cursor

Implicit Cursor and Explicit Cursor.

31. Parameterized Cursor

Cursor accepting parameters.

32. Trigger

Automatically executes on INSERT/UPDATE/DELETE.

33. Types of Trigger

BEFORE Trigger and AFTER Trigger.

34. OLD vs NEW

OLD = previous value, NEW = updated value.

35. Stored Procedure

Precompiled SQL code stored in database.

36. Function

Stored program returning value.

37. Procedure vs Function

Procedure may not return value; function always returns value.

38. DELETE vs DROP

DELETE removes rows; DROP removes table.

39. Aggregation

Combines multiple values into one result.

40. Aggregation Functions

COUNT, SUM, AVG, MAX, MIN.

41. Indexing

Improves searching speed.

42. Aggregation vs Indexing

Aggregation summarizes data; indexing speeds searching.

43. insertOne vs insertMany

insertOne inserts single document; insertMany inserts multiple documents.

44. DBMS vs RDBMS

DBMS stores data generally; RDBMS stores relational tabular data.

45. File in DBMS

Collection of related records stored together.

Important Syntax Examples

Operation	Syntax
Create Table	CREATE TABLE student(id INT, name VARCHAR(20));
Insert	INSERT INTO student VALUES(1,'Ved');
Select	SELECT * FROM student;
Update	UPDATE student SET marks=90 WHERE id=1;
Delete	DELETE FROM student WHERE id=1;
Drop	DROP TABLE student;
Create Index	CREATE INDEX idx_name ON students(name);
MongoDB insertOne	db.users.insertOne({name:'Ved'})
MongoDB insertMany	db.users.insertMany([{name:'Ved'}])
MongoDB find	db.users.find()

Platforms to Run Codes

Technology	Where to Run
MySQL	Terminal / MySQL Shell
MongoDB	mongosh
PL/SQL	Oracle SQL Developer